

MINN2024A PROJECT BLASTING DESIGN APP

Ngwako Raphiri , Thipana Tebogo Rabapane ,
Shabangu Fortune and Mathedimosa Xikombelo

July 2026

1 Abstract

This report develops the design of two calculators, namely the surface blasting design calculator and the fragmentation calculator. These calculators assist blasting engineers by reducing manual calculations and optimizing the blast design calculation process. The calculators calculate parameters such as burden, actual powder factor and mean fragmentation size. The study involved developing algorithms, flowcharts and pseudocodes for the calculators. To improve reliability, input validation was incorporated to ensure that invalid values are not accepted during calculations. The results show that the functions perform the required calculations and generates accurate outputs based on the user provided inputs. Overall, this project demonstrates how technology can be applied to solve practical engineering problems in the mining industry. Furthermore, the project provides strong foundation for future developments such as integrating blasting calculators to further enhance blasting performance and operational efficiency.

2 Background

Blasting is one of the activities that plays a vital role in rock fragmentation in mining. In both surface and underground mining blasting and drilling are used to break down large rocks into manageable sizes for excavation, loading, transportation and mineral processing. To ensure a successful operation, blast engineers are required to develop a proper blast design and perform calculations using parameters such as bench and stemming length, burden, spacing, sub-drill and hole diameter. This process can be time consuming and human errors are a constant threat, especially when multiple blast designs need to be evaluated under different geological and environmental conditions. With the advancing technology solutions are created to automate calculations using algorithms and codes. This enables engineers to perform calculations quickly, providing instant result in the field while improving accuracy. The calculators serve as digital

tools to automate an estimate blasting calculations such as stemming length, burden and mean fragmentation tools. This tool has potential to reduce paperwork while increasing productivity and supporting better decision making in the mining industry.[1]

3 Literature review

Surface Blasting is an important process used to break rocks into smaller manageable pieces which are easier to load. A successful blast depends on the parameters used such as Burden and sub drill these parameters help to improve efficiency and reduce costs while maintaining a safe working environment. Engineers often perform these calculations manually which takes time, the app reduces human error while significantly reducing the time it takes to do the calculations. The app makes blast designing faster more accurate and easier to use[2].

4 The Design

The application was developed using Python. The application requests the user to insert values, and it checks if values are positive or not, it only performs calculations when the conditions are met. The program performs the calculations based on the following sequences: 1.Asks the user to input parameters 2.Checks if the values are not negative 3.It calculates the stemming length 4.Calculates charge length 5.Calculate burden 6.Determine sub-drill based on the rock type 7.Calculate Spacing 8.Calculate the actual powder factor 9.Calculate the mean fragmentation 10.Display all calculated results size of the blasted rock

5 Results Discussion

The program combines multiple blasting calculations into one program, and it works. The conditions given to the inputs prevent the user from entering negative values which reduces errors and gives reliably accurate results. If statements allow the program to calculate different values for Sub-drill and powder factor depending on whether the rock is hard or soft. An issue is that the program asks user to insert results they may have inserted before, this can be improved by storing the original inputs and reusing them and also include more rock types, an option that automatically selects the rock factor based on the rock type inputs[3].

6 Conclusion

The project successfully developed a surface blasting design calculator and fragmentation calculator. The calculators automate calculations that are usually

performed manually, making blast design faster. Input validation implemented in the program ensures that only valid data is processed, to bring about accurate outputs. Beyond achieving the project objectives, the calculators highlight the value of automation and digital solutions in modern mining engineering. Therefore, this project serves not only as a functional calculation tool but also as a step toward digital transformation of blasting operations in mines, with the intent to improve safety and productivity, optimize processes and overall mining performance.[4]

7 Future Work

- Fly-rock prediction and safety zone estimation.
- A database containing different explosive types and their properties.
- Addition of blast vibration and air overpressure prediction calculators.
- Graphical visualization of blast patterns and fragmentation results.

References

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